Financial Modeling

Class 5 — Discretionary Equity Investing

In-class session

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Where we are in the course

- The **strategies** block begins. The construction models determine the size of the positions given the alphas; the strategies are the sources of the alphas
- Round 1 of the AMC is under way. Round 2, the discretionary round, follows, and today's class provides its foundation
- The block continues with quantitative investing and closes with global macro

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Today: the sources of alphas in discretionary investing

- The oldest approach to generating alphas is fundamental analysis, codified by Graham and Dodd in *Security Analysis* (1934): understanding the business, estimating its value, and buying below that value
- A low price has two possible explanations, mispricing or a justified decline in value, and distinguishing between them is the central task of the analyst
- A correct valuation is rewarded only when the market price converges to the estimated value; without a catalyst, the abnormal return is small
- The class closes with an assessment of the approach: what Warren Buffett's track record shows about the returns to value and quality investing

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The discretionary investment process



This is the investment process of a fundamental equity manager, from Graham and Dodd (1934) to the manager of today's case study.

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Exercise: valuing Coca-Cola

Pull the inputs from `finance.yahoo.com` into the yellow cells of sheet *Coca-Cola* (column C says where each one is):

1. quote page of **KO**: the price; *Forward Dividend* (the dollar amount); *Beta (5Y Monthly)*
2. tab *Analysis* of KO: *Growth Estimates*, row *Next Year* (the g_1 of years 1 to 10); *Earnings Estimate*, next year (the EPS, for Part D)
3. quote page of **^TNX**: the 10-year Treasury yield. Given: equity premium 4.2% (Damodaran's implied premium, October 2026); $g_2 = 4\%$ from year 11

Compute, with formulas: the CAPM cost of equity k and the Gordon value $D_0(1+g_1)/(k-g_1)$; then the two-stage value: the dividends of years 1 to 10 at g_1, the terminal value at year 10 from the Gordon formula at g_2, everything discounted at k. **Report** the two-stage intrinsic value per share, in dollars (the green cell).

15 minutes, in pairs.

Then enter your result on Wooclap, before the solution is shown:

`wooclap.com/JAXRAQA`

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Solution: no Gordon value, two-stage value \$120

Cell	Formula (inputs of 26 October 2026)	Value	vs. price
Price P_0	quote page of KO	\$92.04	
Cost of equity k	$4.98\% + 0.32 \times 4.20\%$	6.32%	
Gordon value, $g_1 = 6.70\%$ forever	$2.12 \times 1.0670/(k - 0.0670)$, with $k - g_1 < 0$	none	
PV of the dividends, years 1–10	$\sum_{t=1}^{10} 2.12\,(1.0670)^t/(1+k)^t$	\$21.62	
Terminal value at year 10, $g_2 = 4\%$	$D_{10} \times 1.04/(k - 0.04) = 181.46$, $\times 0.542$	\$98.28	
Two-stage value	$21.62 + 98.28$	**\$119.90**	**+30%**

Terminal growth g_2	3.0%	3.5%	**4.0%**	4.5%	5.0%
Two-stage value	\$89.67	\$102.11	**\$119.90**	\$147.44	\$195.78

- Growth above the cost of equity cannot last: the Gordon formula has no value, the two-stage model does. With $k - g_2$ at 2.3 points, **one point** of terminal growth moves the value from \$90 to \$196. The inputs are estimates: hence the **margin of safety**
- Part D disagrees: at the competitors' forward P/E of 20.2×, Coca-Cola is worth \$71, 23% below the price (the market pays 26.1× its earnings). Each model rests on assumptions that must be stated

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Vote: why catalysts matter

Suppose the two-stage valuation is right and Coca-Cola is worth \$119.90, 30% above the price of \$92.04.

If the mispricing is corrected within one year, the expected abnormal return is 32%.

If it is never fully corrected, the annual alpha is:

about 10% about 3% below 1%

Vote on Wooclap, before the answer is shown: `wooclap.com/JAXRAQA`

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The answer

- **Corrected within a year:** the price moves to next year's value, $P_1 = V_1$, and the dividend is collected. For any dividend discount model (today's value is next year's dividend plus next year's value, discounted once):

$$V_0 = \sum_{t\geq 1} \frac{D_t}{(1+k)^t} = \frac{D_1}{1+k} + \frac{1}{1+k}\sum_{t\geq 2} \frac{D_t}{(1+k)^{t-1}} = \frac{D_1+V_1}{1+k} \;\Rightarrow\; V_1 + D_1 = (1+k)V_0$$

$$\alpha = \frac{V_1+D_1}{P_0} - (1+k) = \frac{(1+k)V_0}{P_0} - (1+k) = (1+k)\frac{V_0-P_0}{P_0} = 1.0632 \times 0.303 = \mathbf{32.2\%}$$

- **Never corrected:** the stock earns a constant expected return $E(R) = k + \alpha$, the rate that discounts the dividends to the *price*: the two-stage formula of the exercise with $E(R)$ in place of k and \$92.04 on the left, solved with Goal Seek, gives $E(R) = 7.01\%$ and

$$\alpha = E(R) - k = 7.01\% - 6.32\% = \mathbf{0.7\%} \text{ per year}$$

- The reward depends on **when** the price converges: 32% within a year, 0.7% a year if never. Hence the guidelines of AMC round 2 require **catalysts that can realize before the end of the challenge**

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Recap: the discretionary strategies and their tools

- **Value investing**: buy below intrinsic value, with a margin of safety
- **Quality investing**, the protection against value traps, to be combined with value: profitability and earnings quality, growth at a reasonable price, safety, payout and management quality
- **Catalyst-driven and activist investing**: identifying, or creating, the event that closes the gap between price and value
- **Valuation tools**: several versions of cash-flow discounting (dividend discount models, single- and multi-stage; the free cash flow model) and relative valuation with comparables, each resting on assumptions that must be stated

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Case study: two cheap stocks

In 2018 the Valuation 50 fund sells Gilead and must buy one health-care stock. Ranked by price over the fund's estimate of intrinsic value, two survive the first cut: **Hologic** (medical devices, $P/V = \mathbf{0.81}$) and **Pfizer** (pharmaceuticals, $P/V = \mathbf{0.67}$). Which one does the manager buy?

$bn	2008	2009	2010	2011	2012	2013	2014	2015	2016	2017	2018
Hologic (fiscal years ending in September)											
Revenue	1.7	1.6	1.7	1.8	2.0	2.5	2.5	2.7	2.8	3.1	3.2
Pre-tax margin (%)	−17	−128	−3	13	−3	−48	2	7	15	40	−13
Free cash flow	0.3	0.5	0.4	0.4	0.3	0.4	0.5	0.7	0.7	0.0	0.7
Acquisitions (cash paid)	2.6	0.0	0.1	0.2	3.8	0.0	0.0	0.0	0.0	1.6	0.1
Impairments of goodwill, intangibles	0.0	2.3	0.2	0.0	0.0	1.1	0.0	0.0	0.0	0.0	0.7
Net debt	0.4	−0.1	0.9	0.8	4.7	4.1	3.5	3.1	2.8	2.8	2.6
Net debt / free cash flow	1.2	–	2.2	1.8	13.8	9.3	7.6	4.3	3.8	–	3.9
Share buybacks	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.2	0.2	0.3
Pfizer (calendar years)											
Revenue	48.3	50.0	67.8	67.4	59.0	51.6	49.6	48.9	52.8	52.5	53.6
Pre-tax margin (%)	20	21	14	19	20	30	25	18	16	23	22
Free cash flow	16.5	15.4	9.9	18.6	15.7	16.6	15.7	13.1	14.1	14.5	13.8
Acquisitions (cash paid)	1.2	43.1	0.3	3.3	1.1	0.0	0.2	16.5	18.4	1.0	0.0
Impairments of goodwill, intangibles	0.0	0.0	3.5	1.2	1.3	1.4	0.5	0.8	1.4	0.4	3.1
Net debt	−6.4	22.7	16.0	12.2	4.8	4.1	0.6	15.7	24.2	23.5	22.9
Net debt / free cash flow	–	1.5	1.6	0.7	0.3	0.2	0.0	1.2	1.7	1.6	1.7
Share buybacks	0.5	0.0	1.0	9.0	8.2	16.3	5.0	5.2	5.0	5.0	12.2

Pre-tax margin: income before income taxes over revenue. Net debt: debt less cash and short-term investments. Hologic 2017: sale of the blood-screening business (pre-tax gain 0.9, tax paid 1.0). Source: 10-K filings, as first reported.

Hologic, Pfizer or neither? Vote on Wooclap, before the answer is shown:

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The answer: underpricing is not sufficient

- The manager buys **Pfizer**, which passes the quality screen: a pre-tax margin between 14% and 31% in every year; free cash flow of 10 to 19 billion a year, with net debt below two years of it; write-offs below 6% of revenue; 67 billion returned through buybacks since 2008
- **Hologic fails the screen** on all three counts: a pre-tax margin negative in five of the eleven years (unsustainable profits); 8.3 billion of acquisitions, 4.4 billion of it written off within a few years (poor stewardship of capital); free cash flow of 0.3 to 0.7 billion a year against net debt that reached 4.7 billion after the Gen-Probe deal (excessive risk). A low price with this record is the profile of a value trap
- **A low valuation alone is not a reason to buy**: the quality screen determines the decision, and it is block 3 of the assignment
- **The catalysts** the fund names for Pfizer: a pipeline with 40 products in Phase III or awaiting approval, a strategic review under way, and a long-term target for organic sales growth.

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Exercise: Buffett against the factors

Sheet *Berkshire*: Berkshire Hathaway's monthly returns since 1985 (BRK-A, Yahoo Finance) and the factors, in the units of their sources and in decimals: Mkt-RF, SMB, HML and RF from Kenneth French's data library, QMJ (USA) from AQR's data sets.

Compute. A: Berkshire's annualised return against the market's, whole sample, by decade, last ten years. B: with the Regression tool (Data, Data Analysis), regress Berkshire's excess return on the market; on Mkt-RF, SMB, HML; adding QMJ: the intercept ($\times 12$) and its t each time.
Report the outperformance over the last ten years (green cell) and the factor that makes the alpha insignificant.

12 minutes, in pairs.

Enter the two answers on Wooclap, before the solution is shown:
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Solution A: the outperformance has faded

Annualised, monthly data	Months	Berkshire	Market	Difference	Growth of 1
Whole sample, Feb 1985 – Jul 2026	498	17.5%	12.4%	+5.2%	549 vs. 99
1985–1989	59	42.0%	17.7%	+24.3%	6.2 vs. 2.2
1990s	120	21.6%	17.6%	+3.9%	6.5 vs. 5.2
2000s	120	7.6%	1.0%	+6.6%	1.8 vs. 1.0
2010s	120	13.4%	13.7%	−0.2%	3.4 vs. 3.6
2020 – Jul 2026	79	14.4%	15.6%	−1.2%	2.3 vs. 2.5
Last ten years	120	14.3%	15.2%	**−0.9%**	3.5 vs. 4.0

- Over four decades, 17.5% a year against 12.4%: 1 dollar became 549 against 99. Most of the margin was earned before 2000; in the 2000s it came from losing less in a flat decade
- Since 2010 Berkshire has not beaten the market. Size is one reason: a portfolio of a trillion dollars cannot hold the concentrated positions of the 1980s

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Solution B: which factors explain the alpha

Regressors (excess returns)	Alpha, annualised	Mkt-RF	SMB	HML	QMJ
Monthly returns, February 1985 to July 2026 (498 months); t-statistics in parentheses					
Market (CAPM)	8.3%	0.66			
	(2.8)	(12.2)			
Mkt-RF, SMB, HML	6.1%	0.78	−0.47	0.44	
	(2.2)	(15.0)	(−6.1)	(6.0)	
+ QMJ	3.9%	0.87	−0.38	0.46	0.30
	(1.4)	(14.4)	(−4.5)	(6.2)	(2.7)

- Berkshire is a large, value, quality portfolio with a beta below one, every loading significant. The CAPM alpha of 8.3% a year ($t = 2.8$) falls to 6.1% with size and value and loses significance once **quality** is added (3.9%, $t = 1.4$)
- What the factors absorb is not stock-picking skill but the consistent, levered application of value and quality over four decades (Frazzini, Kabiller and Pedersen, 2018). The intercept is what remains once the return of a passive portfolio with the same exposures is taken out

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Assignment: the valuation dossier (AMC round 2)

For each stock that you add to your portfolio in round 2, add to the round-2 report:

1. **The method.** Which valuation approach you used (single-stage DDM, multi-stage DDM, comparables) and why it suits this company; which methods are unsuitable for it and why
2. **The valuation.** The intrinsic value with all inputs stated (k, g, the comparables set), and the resulting margin of safety at the current price
3. **The value-trap check.** The quality screen: sustainability of profits, management's stewardship of capital, and free cash flow generation
4. **The catalyst.** The event expected to close the gap between price and value, and the reasons it can realize before the end of the challenge

Graded as complete or incomplete with the round-2 report. This is the analysis required by the guidelines of round 2, in a fixed structure.

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Next class

- **Quantitative investing, Part I**: value and quality as systematic factor strategies across hundreds of stocks, together with momentum and low beta
- The reading and the exercise workbook are on iCorsi
- Reminder: the round-2 report contains the valuation dossier from today, the alphas page from Class 3, and the performance of round 1. The deadline is in the guidelines of the round

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